

- [SIH26229 — Kabadiwala Connect](#)
 - [Bringing the Informal Collector into the Formal Recycling Chain](#)
 - [1. Problem Statement kya keh raha hai \(Official Brief Summary\)](#)
 - [2. Judges ko kya answer dena hai — “Aapne kya innovate kiya?”](#)
 - [Innovation Pillar 1 — “Weighing nahi, Looking Inside” \(Composition-Aware Valuation\)](#)
 - [Innovation Pillar 2 — Trust Layer \(Traceability + Verified vs Declared\)](#)
 - [Innovation Pillar 3 — Accessibility-first Design \(jo apps miss karte hain\)](#)
 - [Innovation Pillar 4 — Government-grade, not consumer-grade](#)
 - [3. Suggested Tech Stack](#)
 - [4. High-Level System Architecture](#)
 - [5. Core Modules / Feature Breakdown](#)
 - [A. Collector App \(PWA\)](#)
 - [B. Recycler Console \(Next.js\)](#)
 - [C. Admin / Ministry Dashboard \(Next.js\)](#)
 - [D. AI/ML Layer](#)
 - [6. PostgreSQL Schema \(Starting Point\)](#)
 - [7. Execution Plan \(36-48 Hour Hackathon Build\)](#)
 - [8. Team Role Split \(6 members\)](#)
 - [9. Unit Economics \(Judges explicitly demand this — mat bhoolna\)](#)
 - [10. Jury Q&A — Prep karke rakho](#)
 - [11. Demo Flow \(Live Presentation ke liye\)](#)
 - [Bottom line](#)

SIH26229 — Kabadiwala Connect

Bringing the Informal Collector into the Formal Recycling Chain

Team Execution & Technical Documentation

1. Problem Statement kya keh raha hai (Official Brief Summary)

Field	Detail
PS Number	SIH26229
Title	Kabadiwala Connect — Bringing the Informal Collector into the Formal Recycling Chain
Organization	Ministry of Mines (MoM) / JNARDDC (Jawaharlal Nehru Aluminium Research Development and Design Centre)
Category	Software
Theme	Clean & Green Technology
Deadline	30 September 2026

Background (asli problem): India apne end-of-life electronics ka bahut bada hissa **informal kabadiwalas / waste-pickers** ke through collect karwata hai — kyunki unka last-mile reach sabse sasta aur widespread hai. Par ye log formal recycling ecosystem (E-Waste Management Rules 2022 ka EPR framework) se bahar hain. Isliye:

- E-waste **cable burning, acid leaching (PCB ko tezaab me daal ke), manual desoldering** jaise unsafe tareeko se process hota hai.
- Copper aur thoda gold nikal liya jata hai, lekin **lithium, cobalt, neodymium, tantalum, gallium, indium** jaise critical/rare minerals **waste ho jaate hain** — jinhe India 90-100% import karta hai.
- Waste-pickers ki health aur ground water dono ko nuksan hota hai.
- Root cause sirf technology ki kami nahi — **information aur institutional gap** hai: collector ko fair price pata nahi, kaunsa recycler authorized hai pata nahi, compliant handover kaise kare pata nahi, transaction ka record nahi milta.

Core ask (Problem statement ka direct maang): Ek **vernacular, low-literacy-friendly, offline-tolerant mobile platform** banao jo: 1. Collector ko fair price discovery de 2. Authorized recyclers se directly connect kare 3. Documented, traceable material handover kare 4. Payment aur earnings ledger maintain kare 5. Formal route ko **economically attractive** banaye — extra compliance burden nahi

2. Judges ko kya answer dena hai — “Aapne kya innovate kiya?”

Market me already Attero MetalMandi, ScrapUncle, The Kabadiwala jaise apps hain — jo generic scrap pricing aur doorstep pickup dete hain. Isliye **sirf ek “scrap pickup app” banana innovation nahi hai**. Judges yahi test karenge. Hamara differentiation ye hona chahiye:

Innovation Pillar 1 — “Weighing nahi, Looking Inside” (Composition-Aware Valuation)

Existing apps sirf **weight × generic rate** karte hain. Hum ek layer aur add karenge: - Har material category (PCB, battery, motor/magnet, cable, CRT, LCD) ke liye JNARDDC/industry ke known **stoichiometric composition ratios** use karke estimate karo ki is batch me approx kitna **Lithium, Cobalt, Neodymium, Copper** hai. - Collector ko sirf ₹ price nahi, ek **“Recovery Score” (0-100)** bhi dikhao — jo batata hai ki kitne critical minerals bachaye gaye aur kitna CO₂ offset hua. - Ye seedha problem statement ke root cause (critical minerals waste ho rahe hain) ko address karta hai — jo koi existing consumer app nahi karta.

Innovation Pillar 2 — Trust Layer (Traceability + Verified vs Declared)

- Collector jo weight declare karta hai (**Declared Weight**) aur recycler jo physically verify karta hai (**Verified Weight**) — dono alag track hote hain. Final payment sirf Verified Weight pe hota hai — isse dono side (collector aur recycler) ka trust build hota hai aur disputes kam hote hain.
- Har handover ka GPS + timestamp + photo based **digital receipt** banega, jo audit/EPR compliance ke liye ministry/recycler dono use kar sakein.

Innovation Pillar 3 — Accessibility-first Design (jo apps miss karte hain)

- Low-literacy UX**: icon/visual heavy interface, voice-based price board (Hindi/Marathi/English), zero-reading-dependency flows.
- Offline-first**: PWA / app jo bina internet ke bhi lot create, price dekhna, aur handover record kar sake — baad me auto-sync.
- Entry-level Android phones pe chale (low RAM, chhota app size).

Innovation Pillar 4 — Government-grade, not consumer-grade

- Built directly on **Ministry of Mines MSP benchmark price board / CPCB authorized recycler list** — official data source, consumer apps ye nahi karte.
- Cash payment allowed by default (digital payment optional) — kyunki collectors ka existing cash economy disrupt nahi karna.

One-liner for judges: “Existing apps sirf scrap weigh karke paisa dete hain. Hum scrap ke andar chhupe critical minerals (Li, Co, Nd, Cu) ko estimate karte hain, unka Recovery Score dikhte hain, aur Declared vs Verified weight ka trust-layer banake informal collector ko formal EPR chain ka hissa banate hain — bina unki cash-based daily earning disturb kiye.”

3. Suggested Tech Stack

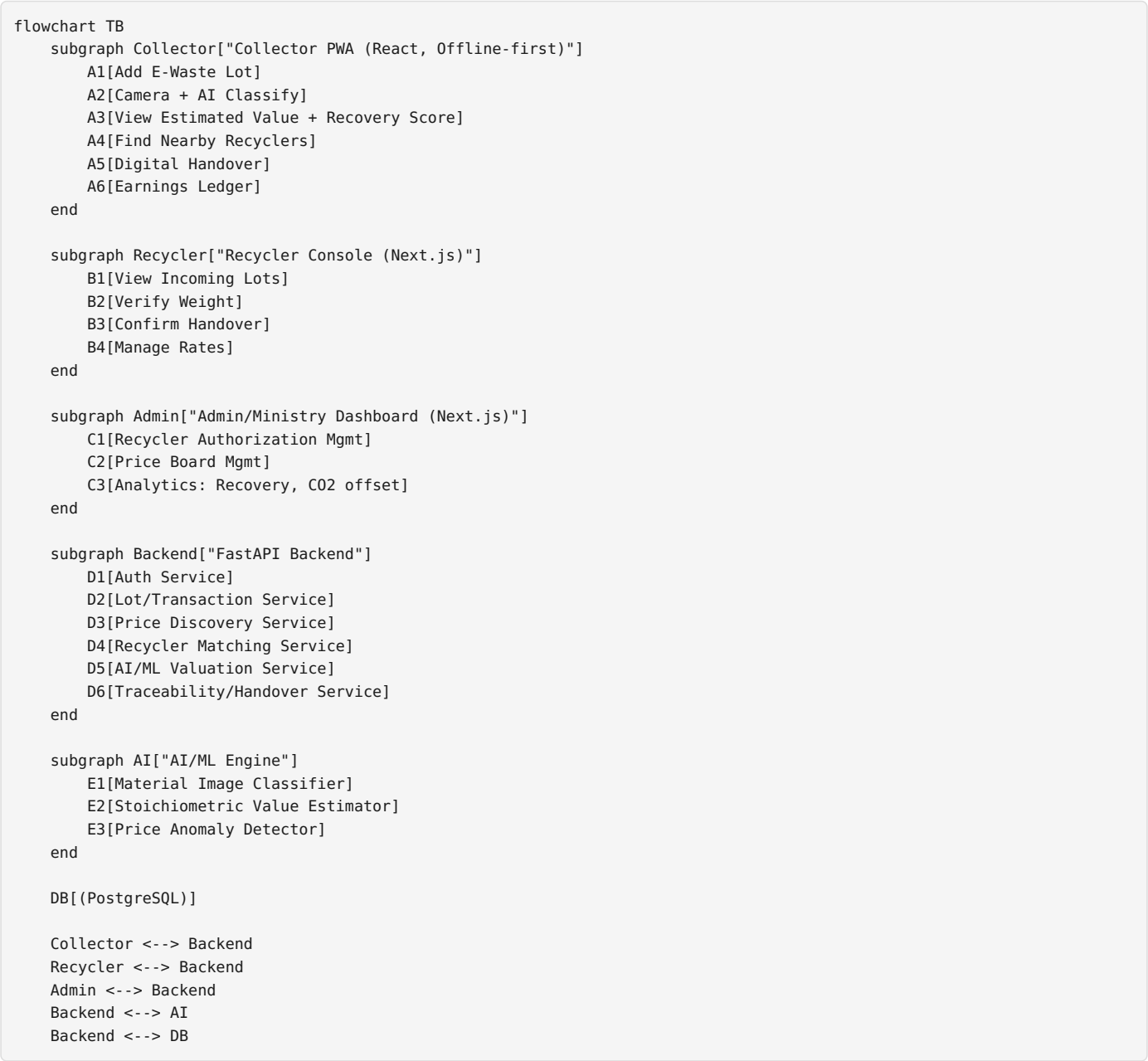
Aapne Python, FastAPI, Next.js, React, PostgreSQL bataya hai — ye sab use hoga, thoda split kar ke:

Layer	Technology	Kyun
Collector Mobile App	React + Vite (PWA) — installable, offline-first via Service Worker + IndexedDB (Dexie.js)	Low-literacy, offline-tolerant, entry-level Android ke liye best; app-store approval ka wait nahi karna
Recycler Console / Admin (Ministry) Dashboard	Next.js + React	SEO/SSR ki zaroorat nahi hai but Next.js ka file-based routing + API routes admin panel ke liye fast dev deta hai
Backend API	FastAPI (Python)	Async, auto Swagger docs (judges ko live API dikhana easy), fast to build in hackathon time
Database	PostgreSQL	Structured relational data (materials, prices, recyclers, transactions) — schema neeche diya hai
AI/ML — Material Classification & Valuation	Python: image classification model (transfer learning — MobileNetV3/EfficientNet-lite, ya multimodal LLM vision API for demo speed) + rule-based stoichiometric calculator	Real-time material category detect karne ke liye
Price Trend / Anomaly Detection	Python (scikit-learn / simple time-series)	Historical price dataset se trend + fraud/abnormal transaction detect karna
Auth	JWT-based auth via FastAPI (OAuth2PasswordBearer)	Collector, Recycler, Admin — 3 roles
Offline Sync	Service Worker + Background Sync API + conflict-resolution queue	Core “offline-tolerant” requirement ke liye mandatory
Voice/Multilingual	Web Speech API (browser-native, free) + i18n (react-i18next) for Hindi/Marathi/English	Low cost, no external API dependency needed for demo

Maps/GPS	Leaflet.js + OpenStreetMap (free, no API key headache)	Recycler discovery + GPS handover proof
Hosting (for demo)	Vercel (Next.js + PWA) + Railway/Render (FastAPI + Postgres)	Fast deploy for judges to test live
Notifications	Web Push API	SMS costs money — Web Push free hai for demo

Tip: Judges FastAPI ka `/docs` (Swagger UI) dekhna pasand karte hain — live API testing dikhana strong impression deta hai.

4. High-Level System Architecture



5. Core Modules / Feature Breakdown

A. Collector App (PWA)

- Onboarding** — minimal profile (name/phone optional, preferred language, general location). PII minimal rakho (problem statement explicitly bolta hai “avoid unnecessary personal info”).
- Add Lot** — photo click karo → AI category suggest karta hai (CRT, LCD, PCB, cable, battery, motor/magnet, mixed plastic) → approx weight enter karo.
- Instant Valuation** — Cash Price (₹) + Recovery Score (0–100) dikhao.
- Price Board** — voice-enabled, current rates by category/location, simple trend arrows (up/down).

5. **Recycler Matching** — location + material + rate + authorization status ke basis pe ranked list.
6. **Digital Handover** — photo + GPS + timestamp + unique reference ID generate; recycler confirm kare.
7. **Earnings Ledger** — transaction history, pending dues, total earned.
8. **Safety Guidance** — pictorial/audio: mat jalao cables, acid mat use karo, battery/CRT safe handling.
9. **Offline mode** — sab upar wale steps bina internet ke locally save ho, connectivity aane pe sync.

B. Recycler Console (Next.js)

1. Incoming lots dashboard
2. Verified Weight entry (declared vs verified — core trust logic yahin hai)
3. Confirm handover → digital receipt auto-generate
4. Manage material rates (jo collector app me reflect ho)
5. Transaction history

C. Admin / Ministry Dashboard (Next.js)

1. Recycler authorization management (CPCB registration verify/approve)
2. Price board oversight (MSP benchmark set karna)
3. Analytics: total material collected, critical minerals recovered, CO₂ offset, active collectors/recyclers, district-wise heatmap

D. AI/ML Layer

1. **Material Classifier** — photo se category detect (transfer learning model / vision API for hackathon speed)
2. **Stoichiometric Valuation Engine** — category + weight → rule-based formula se estimated Li/Co/Nd/Cu content nikalo → Recovery Score calculate karo
3. **Anomaly Detection** — agar quoted price historical range se bahut off hai to flag karo (fraud prevention)
4. **Recycler Recommendation** — location + rate + capacity ke basis pe ranking model

6. PostgreSQL Schema (Starting Point)

```
-- Collectors
CREATE TABLE collectors (
  collector_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  preferred_language VARCHAR(10) DEFAULT 'hi',
  phone VARCHAR(15),
  operating_location GEOGRAPHY(POINT),
  created_at TIMESTAMP DEFAULT now()
);

-- Recyclers
CREATE TABLE recyclers (
  recycler_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  name VARCHAR(150),
  facility_location GEOGRAPHY(POINT),
  materials_accepted TEXT[],
  authorization_status VARCHAR(20), -- 'authorized' / 'pending' / 'unverified'
  registration_number VARCHAR(100),
  contact_details VARCHAR(100),
  service_area_radius_km NUMERIC,
  pickup_available BOOLEAN DEFAULT false
);

-- Material Categories / Price Reference
CREATE TABLE material_prices (
  price_id SERIAL PRIMARY KEY,
  material_category VARCHAR(50),
  sub_category VARCHAR(50),
  location VARCHAR(100),
  buying_price NUMERIC,
  unit VARCHAR(10),
  recycler_id UUID REFERENCES recyclers(recycler_id),
  recorded_at TIMESTAMP DEFAULT now()
);

-- Lots (Material batches)
CREATE TABLE lots (
  lot_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  collector_id UUID REFERENCES collectors(collector_id),
  material_category VARCHAR(50),
  image_ref TEXT,
  declared_weight NUMERIC,
  verified_weight NUMERIC,
  estimated_value NUMERIC,
```

```

recovery_score NUMERIC,           -- 0-100
quoted_price NUMERIC,
final_price NUMERIC,
recycler_id UUID REFERENCES recyclers(recycler_id),
collection_location GEOGRAPHY(POINT),
status VARCHAR(20) DEFAULT 'created', -- created/matched/handover/verified/paid
created_at TIMESTAMP DEFAULT now()
);

-- Traceability / Handover records
CREATE TABLE handovers (
  handover_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  lot_id UUID REFERENCES lots(lot_id),
  photo_ref TEXT,
  gps_location GEOGRAPHY(POINT),
  timestamp TIMESTAMP DEFAULT now(),
  reference_number VARCHAR(50) UNIQUE,
  recycler_confirmation BOOLEAN DEFAULT false
);

-- Transactions / Earnings ledger
CREATE TABLE transactions (
  transaction_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  lot_id UUID REFERENCES lots(lot_id),
  collector_id UUID REFERENCES collectors(collector_id),
  amount NUMERIC,
  payment_mode VARCHAR(10), -- 'cash'/'upi'
  payment_status VARCHAR(20), -- 'pending'/'paid'
  created_at TIMESTAMP DEFAULT now()
);

```

7. Execution Plan (36-48 Hour Hackathon Build)

Phase	Time	Kya karna hai
Idea & Setup	0-4 hrs	Repo setup (frontend/backend split), DB schema finalize, API contract (Swagger) decide, UI wireframes (Figma/paper)
Prototype	4-16 hrs	FastAPI CRUD APIs (lots, recyclers, prices) + Collector PWA basic flow (add lot, view price) + Postgres connected
Integration	16-28 hrs	AI classifier integrate karo, Recovery Score logic, Recycler matching, Handover + GPS, Next.js Recycler & Admin dashboard
Offline & Polish	28-38 hrs	Service Worker + offline sync, multilingual (Hindi/Marathi), voice price board, safety guidance screens, UI polish
Testing + Field Data	38-44 hrs	Kam se kam 2 real/simulated collector interviews (PS explicitly maangta hai — “field research involving at least two working scrap collectors”)
Pitch Prep	44-48 hrs	Demo script, PPT (6-slide AICTE template), unit-economics slide, live demo rehearsal

8. Team Role Split (6 members)

- Backend Lead** — FastAPI, Postgres schema, auth, APIs
- Collector PWA Dev** — React + offline architecture (Service Worker/IndexedDB) + multilingual/voice
- Recycler + Admin Dashboard Dev** — Next.js dashboards
- AI/ML Dev** — image classifier + stoichiometric valuation + anomaly detection
- Integration/DevOps + GIS** — recycler matching, maps, deployment (Vercel + Railway)
- Research + Pitch Lead** — field interviews (mandatory requirement), unit-economics analysis, PPT, demo script

9. Unit Economics (Judges explicitly demand this — mat bhoolna)

PS statement clearly kehta hai: “a short unit-economics assessment comparing the collector’s existing earnings with the potential earnings through the proposed platform”. Ek simple slide banao:

- **Aaj:** Collector ko middleman/local kabadi rate milta hai — approx ₹X/kg copper, PCB ka koi differentiated value nahi milta.
- **Platform ke saath:** Recovery Score-based fair pricing + direct authorized recycler access (middleman cut kam) → estimated ₹X+Y/kg realized value.
- **Platform sustainability:** Small transaction fee from recycler side (recycler ko verified, EPR-compliant supply milta hai — unke liye bhi value hai) ya ministry-funded model (CSR/EPR fund).

10. Jury Q&A — Prep karke rakho

Sawal	Best Answer
“Ye toh already market me hai (Attero, ScrapUncle)?”	Wo consumer-facing generic weight-based apps hain. Hamara focus critical mineral recovery + government EPR chain compliance + offline low-literacy design hai — jo unme nahi hai.
“Real collector se baat ki?”	Haan, field research kiya (kam se kam 2 collectors) — unki actual pain points quote karo.
“Offline kaise kaam karega?”	Service Worker + IndexedDB local storage, background sync jab network aaye, conflict resolution logic dikhao.
“AI ka accuracy kitna reliable hai for critical minerals?”	Clearly bolo ki ye ek estimate/approximation hai based on known stoichiometric ratios, exact lab assay nahi — aur is disclaimer ko app me bhi dikhao (honesty judges ko acchi lagti hai).
“Isse pehle kisi ne solve kyu nahi kiya?”	Connectivity, incentive misalignment (collector ko formal route me fayda nahi dikhta), aur koi bhi kisi structured dataset ka owner nahi hai — ye teeno gaps hum address kar rahe hain.

11. Demo Flow (Live Presentation ke liye)

1. Collector app kholo → offline mode ON karo → lot add karo (photo + AI classify) → Recovery Score dikhao.
2. Nearby authorized recycler dikhao (map + ranking).
3. Handover simulate karo (declared vs verified weight difference dikhao).
4. Recycler console me confirm karo → earnings ledger update hote dikhao.
5. Admin dashboard: total recovered minerals + CO₂ offset chart dikhao.
6. Offline → online sync live demo karo (network toggle off/on karke).

Bottom line

Ye problem statement judges ke liye “yet another scrap app” lag sakta hai agar aap sirf CRUD app bana do. Innovation **Recovery Score + stoichiometric critical-mineral estimation + declared-vs-verified trust layer + genuine offline/low-literacy design** me hai — isi ko baar baar highlight karna, PPT me bhi aur demo me bhi.

Good luck team!